

Li-Yin Young

Superior, CO | email: lilyyoung1122@gmail.com | website: liyo6397.github.io

Skills

Languages: Python, C, SQL

DevOps & CI/CD: GitHub Actions, GitLab CI/CD, cibuildwheel

Platforms & Tools: GCP, Git, Linux, Bash/Shell scripting, CPython

Experience

Cooperative Institute for Research in Environmental Sciences

Boulder, CO

Software Engineer

April 2021 - Present

- Mentored 3 software engineers through the full lifecycle of multiple open-source Python packages and web APIs, and documented the release process for the broader team.
- Partnered with the science team to integrate and deploy 5 scientific models in a Python package published on PyPI, with individual packages downloaded 7,000-52,000 times.
- Owned repository guardrails, CI/CD criteria, and final PR review across multiple GitHub and GitLab repositories as the primary merge gate for incoming changes.
- Built and shipped cross-platform Python wheels for C and Fortran backends using CPython, cibuildwheel, and GitHub Actions, reducing build time by 86% through parallelized builds.
- Designed and built a unified Python interface for multiple geomagnetic field models used by NGA, enabling batch processing through a single API and shipping 3 production releases.

Main Street Exchange

Boulder, CO

Full Stack Developer

Jun 2016 - May 2018

- Implemented scripting tools and virtual server environments to troubleshoot real-time system issues.
- Built major portal functionality, including third-party app integration and database management.

Topic Technology

Boulder, CO

Machine Learning Developer

Jan 2016 - May 2016

- Built tools to extract unstructured sentiment from social media for training machine learning models.

Millennium Venture Systems

Colorado Springs, CO

Machine Learning Engineer Internship

Jul 2014 - Aug 2014

- Built a support vector machine application in C++ for time series prediction.

Education

University of Colorado Boulder

2018-2020

Master of Science in Applied Mathematics

Boulder, CO

University of Colorado Boulder

2013-2015

Master of Science in Computer Science

Boulder, CO

Research Publications

- Chulliat, Arnaud et al. (2025). The US/UK World Magnetic Model for 2025-2030: Technical Report. <https://doi.org/10.25923/prbc-s316>
- Nair, M., Redmon, R., Young, L.-Y., Chulliat, A., Trotta, B., Chung, C., et al. (2023). MagNet—A data-science competition to predict disturbance storm-time index (Dst) from solar wind data. *Space Weather*, 21, e2023SW003514. <https://doi.org/10.1029/2023SW003514>
- Li-Yin Young. 2018. The Effect of Moderator Bots on Abusive Language Use. In *Proceedings of the International Conference on Pattern Recognition and Artificial Intelligence (PRAI 2018)*. Association for Computing Machinery, New York, NY, USA, 133–137. <https://doi.org/10.1145/3243250.3243257>